

Meta-analysis of gene expression profiles to predict response to biologic agents in rheumatoid arthritis

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Abstract Our aim was to identify differentially expressed (DE) genes and biological processes that may help predict patient response to biologic agents for rheumatoid arthritis (RA). Using the INMEX (integrative meta-analysis of expression data) software tool, we performed a meta-analysis of publicly available microarray Gene Expression Omnibus (GEO) datasets that examined patient response to biologic therapy for RA. Three GEO datasets, containing 79 responders and 34 non-responders, were included in the meta-analysis. We identified 1,374 genes that were consistently differentially expressed in responders vs. non-responders (651 up-regulated and 723 down-regulated). The up-regulated gene with the smallest p value ($p=0.000192$) was *ASCC2* (Activating Signal Cointegrator 1 Complex Subunit 2), and the up-regulated gene with the largest fold change (average log fold change = -0.75869 , $p=0.000206$) was *KLRC3* (Killer Cell Lectin-Like Receptor Subfamily C, Member 3). The down-regulated gene with the smallest p value ($p=0.000195$) was *MPL* (Myeloproliferative Leukemia Virus Oncogene). Among the 236 GO terms associated with the set of DE genes, the most significantly enriched was “CTP biosynthetic process” (GO:0006241; $p=0.000454$). Our meta-analysis identified genes that were consistently DE in responders vs. non-responders, as well as

biological pathways associated with this set of genes. These results provide insight into the molecular mechanisms underlying responsiveness to biologic therapy for RA.

Keywords Biologic agent · Gene expression · Meta-analysis · Response · Rheumatoid arthritis

Introduction

Rheumatoid arthritis (RA) is a chronic inflammatory disease that predominantly affects synovial joints. The disease affects an estimated 1 % of the world's population [1]. Tumor necrosis factor- α (TNF- α) is a potent pro-inflammatory cytokine that plays an important role in inflammatory and immune responses, including those involved in the pathogenesis of RA [2, 3]. Evidence of the central role played by TNF in RA led to the introduction of anti-TNF therapy. Rituximab is a chimeric monoclonal antibody that targets CD20⁺ B cells [4]. It binds to CD20, and causes depletion of subpopulations of peripheral B cells via several putative mechanisms, which include cell-mediated and complement-dependent cytotoxicity, as well as promotion of apoptosis [4, 5].

Biologic agents, such as TNF antagonists and rituximab, are among the most effective therapies for RA. However, not all patients respond. The reasons for this lack of response are unclear, but are the subject of considerable research interest. The ability to predict patient response to biologic agents would facilitate appropriate use of biologic therapy and reduce treatment costs. Overall, this would represent a significant advance in patient care.

The application of high-throughput genomic technologies, such as microarrays, has helped improve our understanding of complex interactions and networks in disease development [6, 7]. Identification of genes associated with responder status may lead to the discovery of potential biomarkers for

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predicting patient response to biologic therapy, and provide insights into RA pathogenesis [8]. Successful identification of gene expression signatures that differentiate healthy and diseased states depends upon the size and quality of the samples, and has generally yielded heterogeneous datasets [9]. Although microarray studies generate lists of differentially expressed (DE) genes, findings tend to be inconsistent across studies, because sample sizes are often small [10].

To overcome this limitation, meta-analyses have been applied to publically available genome-wide gene expression datasets relating to complex diseases [11, 12]. Use of meta-analysis can enhance reliability and generalizability of the results, and permit more consistent identification of relevant gene expression markers [13]. Meta-analyses can provide enhanced statistical power, allowing the identification of more robust and reliable gene signatures [7, 14–16]. To overcome the limitations of individual studies, resolve inconsistencies, and reduce the rate of false-positive and false-negative associations generated by random errors, we turned to meta-analysis. The objective of this study was to identify DE genes and biological processes associated with favorable response to biologic RA therapy, using a microarray meta-analysis approach.

Materials and methods

Identification of eligible gene expression datasets of RA

Searches for microarray datasets that examined DE genes in responders and non-responders to biologic RA therapy were conducted throughout July 2013. To identify suitable microarray datasets, we used the NCBI GEO (Gene Expression Omnibus) database (<http://www.ncbi.nlm.nih.gov/geo/>) [17]. The keyword “rheumatoid arthritis” was used for the search. A study was included in the analysis if: (1) it contained gene expression profiling of blood or synovium before biologic therapy, (2) it provided sufficient data to perform meta-analysis, and (3) it included patients diagnosed with RA based on the American Rheumatism Association classification criteria [18]. The following information was extracted from each study selected: (1) GEO accession, (2) sample type, (3) platform, (4) numbers of responders and non-responders, and (5) gene expression data.

Meta-analysis of microarray datasets

All available microarray datasets fulfilling the inclusion criteria were downloaded from the NCBI GEO database. We constructed data tables containing gene expression or relative gene expression values, with genes/probes in rows and samples/experiments in columns. After uploading the datasets to INMEX (<http://www.inmex.ca/INMEX>), we annotated

them by converting different gene or probe IDs to Entrez IDs. We inspected the data to determine whether the class labels were consistently used across different datasets. For each probeset, intensity values were log-transformed and/or normalized to zero mean and unit variance [19]. After all datasets were uploaded, processed, and annotated, we performed a data integrity check before proceeding to the meta-analysis stage.

One disadvantage of combining p values or standardized differences is that the results can be affected by outliers, which are common in microarray data. A method for combining rank statistics can be used to address this problem, since non-parametric approaches based on rank orders are more robust for such data. We therefore applied a meta-analysis method based on combining rank orders. The implementation is based on the RankProd package, as described by Hong et al. [20]. Briefly, instead of p values or effect sizes, the ranks of DE evidence are calculated for each gene in each study. For each data set, the fold changes (FC) are computed for all possible pairwise comparisons. The ranks of the FCs within each comparison are then used to calculate the rank product for each gene. Permutation tests are then performed to assess the null distributions of the rank products within each data set. The whole process is repeated multiple times to compute the p value and false discovery rate associated with each gene. Statistical analyses were performed using the INMEX program [21].

Functional analysis

The functional analysis module of INMEX is designed to generate new hypotheses by exploiting characteristics of the DE gene lists identified in the previous meta-analysis. The “Pattern extractor” tool was used to visualize gene expression profiles across different datasets/conditions as heat maps. To further understand the functions of the subset of genes identified, we performed Gene Ontology (GO) enrichment analysis, using the hypergeometric test to detect enriched functional attributes based on gene-associated GO terms (<http://www.geneontology.org/>) [22].

Results

Studies included in the meta-analysis

Three GEO data sets were found to meet our inclusion criteria (Table 1), namely two blood datasets and one synovium dataset, obtained before biologic therapy, and included a total of 79 responders and 34 non-responders. Rituximab was used in two of the studies, and TNF inhibitors in the third. Follow-up periods ranged from 12 weeks to 6 months. Two of the studies used the Affymetrix Human Genome U133 Plus 2.0

Table 1 Characteristics of the individual studies included in the meta-analysis

GEO accession	Numbers		Sample	Biologic agent	Response criteria	Follow-up period	Platform
	R	NR					
GSE15258	53	22	Blood	TNF inhibitors	EULAR response	14 weeks	GPL570, Affymetrix Human Genome U133 Plus 2.0 Array
GSE24742	18	6	Synovium	Rituximab	EULAR response	12 weeks	GPL570, Affymetrix Human Genome U133 Plus 2.0 Array
GSE37107	8	6	Blood	Rituximab	EULAR response	6 months	GPL6947, Illumina HumanHT-12 V3.0 Expression BeadChip

Ref reference, GEO Gene Expression Omnibus, GSE Gene Expression Series, GPL Gene Platform, R responder, NR non-responder

Array; the third used the Illumina HumanHT-12 v3.0 Expression BeadChip. Selected details of the individual studies are summarized in Table 1. Heat maps, based on DE analysis of

individual data sets, were used to visualize changes in expression patterns for a subset of genes from the three studies (Fig. 1).

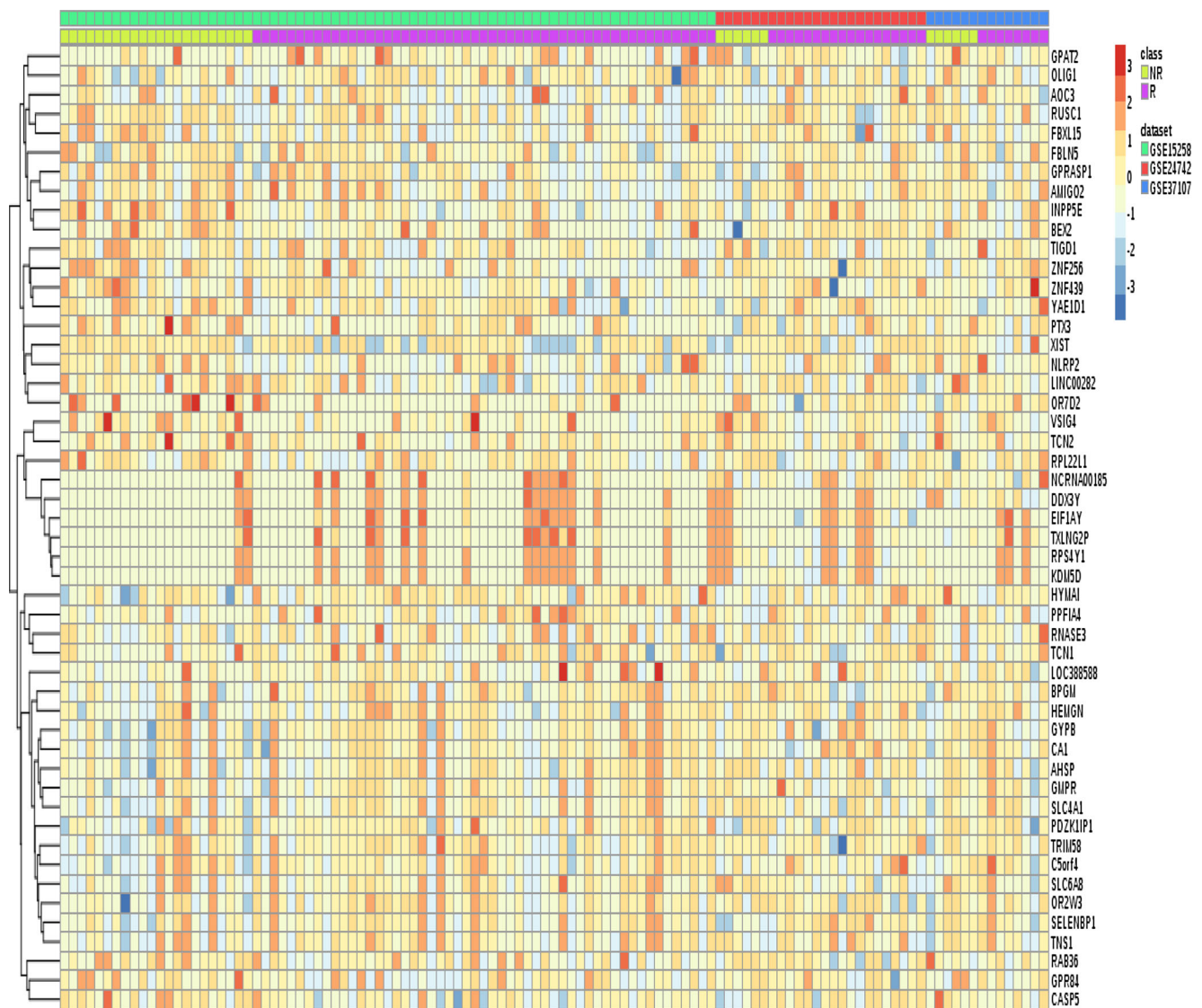


Fig. 1 Heat map visualization of the patterns of change for a particular gene across different datasets (row-wise comparison). For visualization, individual datasets were re-scaled in order to prevent the view from being dominated by study-specific effects

Table 2 The 10 most strongly up-regulated genes in responders vs. non-responders

Entrez ID	Gene symbol	Average log FC	<i>p</i> value	Gene name
84164	<i>ASCC2</i>	−0.15733	0.000192	Activating Signal Cointegrator 1 Complex Subunit 2
8395	<i>PIP5K1B</i>	−0.13898	0.000193	Phosphatidylinositol-4-phosphate 5-kinase, type I, Beta
220359	<i>TIGD3</i>	−0.11797	0.000195	Tigger Transposable Element Derived 3
8814	<i>CDKL1</i>	−0.18779	0.000202	Cyclin-Dependent Kinase-Like 1 (CDC2-Related Kinase)
403340	<i>MGC70870</i>	−0.18516	0.000202	C-terminal binding protein 2 pseudogene
440712	<i>C1orf186</i>	−0.4922	0.000203	Chromosome 1 Open Reading Frame 186
3823	<i>KLRC3</i>	−0.75869	0.000205	Killer Cell Lectin-Like Receptor Subfamily C, Member 3
130355	<i>C2orf76</i>	−0.03186	0.000206	Chromosome 2 Open Reading Frame 76
9717	<i>SEC14L5</i>	−0.09321	0.000207	SEC14-Like 5 (<i>S. cerevisiae</i>)
79618	<i>HMBOX1</i>	−0.51857	0.000208	Homeobox Containing 1

FC fold change

Meta-analysis of gene expression patterns in responders vs. non-responders

We selected the DE genes with *p* value <0.05. There were 1,366 “gained” genes and 49 “lost” genes in this meta-analysis. Gained genes are DE genes identified in the meta-analysis, but not in any of the individual studies [16]. These genes showed relatively weak but consistent expression profiles across all three datasets. The increased sample size attained by combining the datasets provides additional confidence for declaring them as DE genes. Lost genes are genes that were identified as differentially expressed in one or more individual analyses, but not in the meta-analysis. These genes either showed conflicting changes in expression profiles, or very large variations across different studies.

Identifying genes differentially expressed in responders vs. non-responders

We identified 1,374 genes across studies that were consistently differentially expressed in responders vs. non-responders. Among the 1,374 DE genes, 651 genes were up-regulated,

and 723 genes were down-regulated. A list of the top 10 most significantly up- or down-regulated genes is presented in Tables 2 and 3. The up-regulated gene with the lowest *p* value (*p*=0.000192) was *ASCC2* (Activating Signal Cointegrator 1 Complex Subunit 2), which is known to play a role in enhancing NF-kappa-B, SRF, and AP1 transactivation [23]. The up-regulated gene with the largest FC (average log FC=−0.75869, *p*=0.000206) was *KLRC3* (Killer Cell Lectin-Like Receptor Subfamily C, Member 3), which encodes a receptor involved in the recognition of MHC class I HLA-E molecules by NK cells and some cytotoxic T cells [24] (Table 2). The down-regulated gene with the smallest *p* value (*p*=0.000195) was *MPL* (Myeloproliferative Leukemia Virus Oncogene). *MPL* functions as a receptor for thrombopoietin [25], and may be a regulatory molecule specific for TPO-R-dependent immune responses (Table 3).

Functional analysis

To identify biological processes associated with gene expression differences in responders vs. non-responders, we

Table 3 The 10 most strongly down-regulated genes in responders vs. non-responders

Entrez ID	Gene symbol	Average log FC	<i>p</i> value	Gene name
4352	<i>MPL</i>	0.026972	0.000195	Myeloproliferative Leukemia Virus Oncogene
124790	<i>HEXIM2</i>	0.00343	0.000196	Hexamethylene Bis-Acetamide Inducible 2
256236	<i>NAPSB</i>	0.017253	0.000198	Napsin B Aspartic Peptidase, Pseudogene
10098	<i>TSPAN5</i>	0.044641	2.00E-04	Tetraspanin 5
643008	<i>C17orf109</i>	0.255307	0.000201	Chromosome 17 Open Reading Frame 109
3145	<i>HMBS</i>	0.059609	0.000204	Hydroxymethylbilane Synthase
55544	<i>RBM38</i>	0.184578	0.000207	RNA Binding Motif Protein 38
219899	<i>TBCEL</i>	0.013345	0.00021	Tubulin Folding Cofactor E-Like
84311	<i>MRPL45</i>	0.01305	0.000215	Mitochondrial Ribosomal Protein L45
400946	<i>FLJ12334</i>	0.309844	0.000218	Uncharacterized LOC400946

FC fold change

performed GO enrichment analysis with the DE genes identified in our meta-analysis. We found 236 instances of significant enrichment (Supplementary data), corresponding to 10 GO terms (Fig. 2). The most significantly enriched GO term was “CTP biosynthetic process” (GO:0006241) ($p=0.000454$). Other significantly enriched GO terms included “pyrimidine ribonucleoside triphosphate biosynthetic process” (GO:0009209) ($p=0.000454$) and “CTP metabolic process” (GO:0046036) ($p=0.000454$;

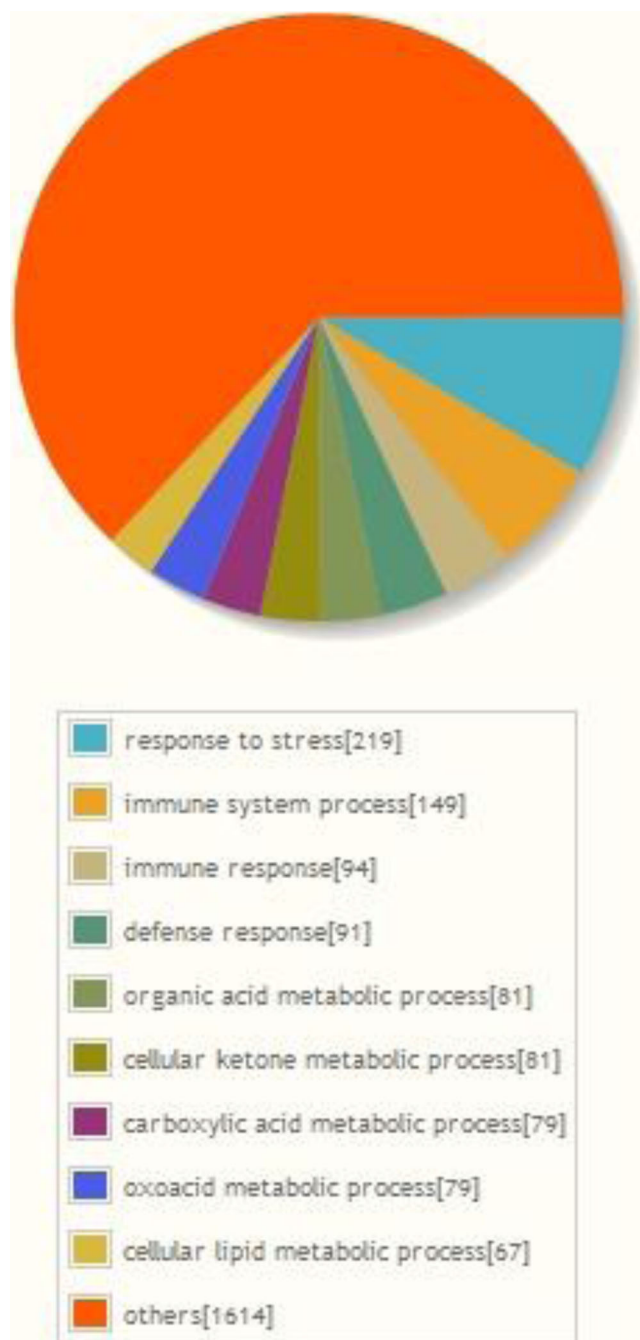


Fig. 2 Summary of the enriched GO terms associated with the list of DE genes in responders vs. non-responders

Table 4). In addition, our analysis suggests that the biological processes represented by the terms “immune system process” (GO:0002376), “negative regulation of glycogen biosynthetic process” (GO:0045719), and “organic acid metabolic process” (GO:0006082), “carboxylic acid metabolic process” (GO:0019752), and “oxoacid metabolic process” (GO:0043436) might be relevant to the response to biologic therapy for RA (Table 4).

Discussion

Many genes are likely to be differentially expressed in responders vs. non-responders to biologic RA therapy. The challenge is to identify subsets of those genes that can help us to understand the molecular and cellular processes which determine response to biologics in RA patients [26]. Although microarray studies produce huge amounts of data, the small sample size of many studies is a major obstacle to the identification of reliable biomarkers. Meta-analysis of multiple microarray datasets can increase the effective sample size, potentially leading to more reliable identification of informative genes and improved predictive power.

Accurate prediction of patient responsiveness to biologic therapy is an important goal in RA research. We therefore sought to identify biomarkers that can discriminate between responders and non-responders. We performed a meta-analysis of three microarray studies that examined responsiveness to biologics in RA patients. Across the three studies, we identified 1,374 genes that were consistently differentially expressed in responders vs. non-responders (651 up-regulated and 724 down-regulated). The up-regulated gene with the smallest p value was *ASCC2*, which enhances transactivation of transcription factors [23]. The up-regulated gene with the largest FC was *KLRC3*, which encodes a receptor for the recognition of MHC class I HLA-E molecules [24]. The down-regulated gene with the smallest p value was *MPL*, which encodes a receptor for thrombopoietin [25]. Although it is currently unclear how these genes might contribute to RA, and interpretation of these results is not straightforward, further studies are warranted because of their potential as predictors of therapy response.

We identified 236 significantly enriched GO terms associated with the DE genes in our meta-analysis. The three most significantly enriched terms were “CTP biosynthetic process”, “pyrimidine ribonucleoside triphosphate biosynthetic process”, and “CTP metabolic process”. In addition, our analysis suggested that processes related to the GO terms “immune system process”, “negative regulation of glycogen biosynthetic process”, “organic acid metabolic process”, “carboxylic acid metabolic process”, and “oxoacid metabolic process”, might also be relevant to responder status. “CTP biosynthetic process” refers to the chemical reactions and pathways leading

Table 4 The top 10 enriched GO terms among the DE genes in responders vs. non-responders

GO ID	Term	<i>p</i> value	Genes
GO:0006241	CTP biosynthetic process	0.000454	NME9; TXNDC3; NME6; NME7; UCK2; CTPS
GO:0009209	Pyrimidine ribonucleoside triphosphate biosynthetic process	0.000454	NME9; TXNDC3; NME6; NME7; UCK2; CTPS
GO:0046036	CTP metabolic process	0.000454	NME9; TXNDC3; NME6; NME7; UCK2; CTPS
GO:0002376	Immune system process	0.000473	AHSP; BPGM; VSIG4; PTX3; FKBP1B; IGF; MMP9; SNCA; ITGAV; PBX1; GATA1; EPB42; LILRA5; ALAS2; DTX1; CARD9; AIM2; VNN1; GAPT; MILR1; SLC7A5; TREML2; KIF13B; F12; CLEC10A; MZB1; UBASH3A; CCL23; PGLYRP1; KLRD1; CACNB3; CD79A; PAXIP1; CTSG; KCNN4; TRIM10; CEBPE; CD19; OAS3; CD1A; PRKX; PSMD3; PLA2G7; CD274; ISG15; THBD; IL18RAP; HERC5; CLU; SOX6; ITGB5; LGALS3; TBX21; BIRC3; AZU1; ITGA4; BST2; CLEC4D; GLMN; ESAM; OSM; C5; NR1H3; IL5RA; CXCL10; MLST8; GF11B; S100B; OASL; PPBP; MX1; FCER2; CDKN2B; KLRG1; CREB3; CD226; SPON2; MEN1; ZBP1; LIG4; HOXA3; IL8; TESC; CD3E; BLK; ENPP2; KIR2DL1; FOS; VASP; NFKBIL1; BNIP3; BCL2L1; IKBKE; MAP2K6; ICAM4; RELB; CD160; CD180; CCR5; BAK1; CD38; ODZ1; GLRX5; FAS; S1PR4; PF4; ATP1B1; G6PD; TAL1; SH2D1B; EMR2; GRAP2; HLA-DQB1; MS4A1; TNFSF4; HSH2D; MARCH8; TRAT1; RPS6KA5; GZMM; AGER; CHID1; PTGDR2; PSMA8; BSG; IFIT3; DHX58; TGFBR3; ADAMDEC1; CRCP; IL18; ROGDI; PTPRJ; LTA; ELMO1; SEC24D; XRCC4; TBKBP1; KIR2DS5; AP2B1; EOMES; TRIM32; SLAMF7; SPIB; IL28RA; CD1C; XCL1; CFD; LYST
GO:0045719	Negative regulation of glycogen biosynthetic process	0.000653	GRB10; INPP5K; GSK3A; PASK
GO:0009208	Pyrimidine ribonucleoside triphosphate metabolic process	0.000661	NME9; TXNDC3; NME6; NME7; UCK2; CTPS
GO:0009148	Pyrimidine nucleoside triphosphate biosynthetic process	0.001286	NME9; TXNDC3; NME6; NME7; UCK2; CTPS
GO:0006082	Organic acid metabolic process	0.001455	C5orf4; SLC6A8; SNCA; IDO1; ABCG2; ALOX15; ALDH5A1; ALOX12; ASL; SARS2; HMGCL; VNN1; LARS2; SLC7A5; ODC1; VARS2; GSTZ1; ADHFE1; ALDH8A1; DHRS9; GSTA1; FAM206A; NARS2; SULT1B1; GGH; PSMD3; AGPAT1; PDHX; OXSM; CYP4F2; TWIST1; GSTM3; NR1H3; GSTM2; RNPEP; FAAH; GADL1; AACs; AS3MT; ACSM2A; LIAS; FADS1; PHYH; ETHE1; POR; GGTLC1; SUOX; VARS; PEMT; CYP27A1; CARNS1; AMT; MCEE; IARS2; PRPS1; G6PD; PCBD1; HPGD; CTPS; CRYL1; ACSL4; GGTA1P; YOD1; PSMA8; PTS; BSG; PFKFB2; HSD3B7; STRADB; ACAD8; PLA2G4A; ACSF2; CYP17A1; EC12; ACOT8; BCKDHA; ABHD5; KMO; SLC27A3; HSD17B8; NAGS
GO:0019752	Carboxylic acid metabolic process	0.001832	C5orf4; SLC6A8; SNCA; IDO1; ALOX15; ALDH5A1; ALOX12; ASL; SARS2; HMGCL; VNN1; LARS2; SLC7A5; ODC1; VARS2; GSTZ1; ADHFE1; ALDH8A1; DHRS9; GSTA1; FAM206A; NARS2; SULT1B1; GGH; PSMD3; AGPAT1; PDHX; OXSM; CYP4F2; TWIST1; GSTM3; NR1H3; GSTM2; RNPEP; FAAH; GADL1; AACs; AS3MT; ACSM2A; LIAS; FADS1; PHYH; ETHE1; POR; GGTLC1; SUOX; VARS; PEMT; CYP27A1; CARNS1; AMT; MCEE; IARS2; G6PD; PCBD1; HPGD; CTPS; CRYL1; ACSL4; GGTA1P; YOD1; PSMA8; PTS; BSG; PFKFB2; HSD3B7; STRADB; ACAD8; PLA2G4A; ACSF2; CYP17A1; EC12; ACOT8; BCKDHA; ABHD5; KMO; SLC27A3; HSD17B8; NAGS
GO:0043436	Oxoacid metabolic process	0.001832	C5orf4; SLC6A8; SNCA; IDO1; ALOX15; ALDH5A1; ALOX12; ASL; SARS2; HMGCL; VNN1; LARS2; SLC7A5; ODC1; VARS2; GSTZ1; ADHFE1; ALDH8A1; DHRS9; GSTA1; FAM206A; NARS2; SULT1B1; GGH; PSMD3; AGPAT1; PDHX; OXSM; CYP4F2; TWIST1; GSTM3; NR1H3; GSTM2; RNPEP; FAAH; GADL1; AACs; AS3MT; ACSM2A; LIAS; FADS1; PHYH; ETHE1; POR; GGTLC1; SUOX; VARS; PEMT; CYP27A1; CARNS1; AMT; MCEE; IARS2; G6PD; PCBD1; HPGD; CTPS; CRYL1; ACSL4; GGTA1P; YOD1; PSMA8; PTS; BSG; PFKFB2; HSD3B7; STRADB; ACAD8; PLA2G4A; ACSF2; CYP17A1; EC12; ACOT8; BCKDHA; ABHD5; KMO; SLC27A3; HSD17B8; NAGS

to the formation of CTP (cytidine 5'-triphosphate), which may regulate microtubule physiology [27]. Because these pathways were consistently enriched across all of the datasets, it

is likely that this represents a biologically relevant finding. Some of these might have interesting functions that could potentially regulate the response to biologics in RA patients.

Some limitations of the present study should be noted. Firstly, heterogeneity and other confounding factors may have distorted the analysis. Clinical samples could be heterogeneous with respect to clinical findings, severity, autoantibody status, or gender. Secondly, there are likely to be basal differences in gene expression between blood and synovium that were not taken into account here. We could not perform subgroup analysis based on tissue type, due to the small number of cases. However, integrating data from different tissues may have enabled us to detect genes that we would have missed in a subgroup analysis. Thirdly, we could not perform subgroup analysis based on the type of biologic agent used (anti-TNF agents or rituximab), due to the small number of cases for some treatment conditions. However, it would be interesting to compare responder vs. non-responder gene signature differences for various biologic agents. Fourthly, current evidence may suggest unexpected “culprits” in disease when a protein encoded for by a suspect gene is not part of the currently identified pathway of pathogenesis. However, the results were quite modest. High-throughput genomic studies have not proven particularly helpful at least in immunology, whereas there are examples of the second strategy being quite useful, for instance, the discovery that a deficiency of functional factor H in the complement system is associated with macular degeneration [28]. This may be largely due to small effect sizes of individual genetic associations.

In conclusion, our meta-analysis of microarray studies provides an overview of differential gene expression signatures in responders vs. non-responders to biologic therapy for RA. One thousand three hundred seventy-four genes were differentially expressed in responders vs. non-responders. Of these, 651 were up-regulated and 723 were down-regulated. Further studies to validate the genes as predictors of response to biologic therapy for RA may provide insight into their roles in development or progression of the disease.

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Conflict of interest statement The authors have no conflict of interest to declare.

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